

Failure Modes and Fair Evaluation of Precision Curricula for In-Hand Reorientation

Abstract—Competence-gated curricula are common in dexterous manipulation, yet their behavior near precision limits remains poorly understood. Curriculum scheduling is studied for three-dimensional in-hand reorientation using a 24-degree-of-freedom Shadow Hand with 20 actuated joints. During training, success tolerance decreases from 0.10 to 0.01 rad, while only the competence-to-tolerance mapping varies. Monotone promotion gates collapse because noisy competence estimates can raise difficulty beyond sustainable levels without a recovery path. Even a denoised gate with dwell buffering collapses in five of ten seeds. Adding demotion prevents collapse in all ten seeds for both Automatic Domain Randomization and variance-aware gates. Bidirectional gates reverse premature promotions and recover repeatedly, improving training reliability without raising the precision ceiling. Evaluation at common tolerances confirms that no method masters 0.01 rad. Across five replications and an estimator noise sweep, the variance-aware gate shows no detectable advantage over the plain bidirectional rule, suggesting that reversibility makes additional caution redundant. Experiments across three simulated hands demonstrate that falling competence ceilings produce both over-promotion collapse and under-promotion stall, making fixed scalar thresholds unreliable across embodiments.

Index Terms—Dexterous manipulation, reinforcement learning, curriculum learning, in-hand reorientation, evaluation methodology.

I. INTRODUCTION

Precision is where curricula earn their keep. Training a multi-fingered hand to reorient an object to within 0.01 rad ($\approx 0.6^\circ$) of a goal orientation fails outright when attempted from scratch, while the same task at a loose tolerance is readily solved. The standard remedy is a competence-gated curriculum: begin at a forgiving tolerance and tighten it as a running estimate of the policy’s skill crosses a promotion threshold [1], [2]. Nearly every published dexterous-manipulation system contains such a gate somewhere in its training stack [3], [4], [5], [6], [7], usually described in a sentence and tuned by hand. Whether the gate itself behaves well when pushed toward a hand’s precision limit has, to our knowledge, not been asked. *Our research objective is to characterize how competence-gated curricula behave at a hand’s precision frontier: to identify the failure modes and their common cause, and to establish a fair protocol for comparing schedulers there.*

The study runs on the Isaac Lab Shadow Hand reorientation task, using the success tolerance τ as a continuous difficulty axis and holding everything else fixed, with the same PPO learner, reward, environments, and frame budget. Only the scheduling rule that maps competence readings to tolerance varies between experimental arms, which makes the comparison causal. GPU-parallel simulation [8], [9]

keeps it affordable, one run taking roughly 36 minutes on a single consumer GPU, so the full study fits in about a week of compute, spanning seven scheduler arms (six at ten seeds on the headline object plus a single-seed fixed clock), three objects (cube, egg, pen), and two further hands (Allegro, Revo3). We report four findings:

- 1) **Failure characterization.** Monotone competence gates, the kind that only ever tighten, collapse at precision. A noisy competence estimate lets a lucky window clear the promotion threshold, difficulty steps beyond what the policy can sustain, the success signal starves, and with no way back the run cannot recover. Training directly at $\tau = 0.01$ collapses in 9 of 10 seeds, and the failure survives de-noising. A fair monotone gate carrying the same smoothing and dwell buffering as the bidirectional arms still collapses in 5 of 10 seeds, so the pathology belongs to monotonicity itself rather than to a weak baseline.
- 2) **A fix from prior art.** A gate that can also step difficulty back down turns collapse into graceful degradation, with zero collapses in 10 seeds and a settled tolerance at the tightest the policy can sustain ($\tau_{\text{final}} \approx 0.020$ – 0.027 rad). Bidirectional pacing is not new. It is the expand-and-contract rule of Automatic Domain Randomization (ADR) [4] moved from a randomization axis to a precision axis, and it serves here as the baseline. The contribution is the demonstration, by ablation and within-run counts, that the demotion branch is the single load-bearing mechanism.
- 3) **A fair ruler.** Arms that settle at different tolerances cannot be compared on their own final scores. A tolerance-matched protocol scores every trained policy at a common fixed grid, and under it no method masters 0.01 rad and the arms are not statistically separable there. Bidirectionality thus contributes reliable training rather than a new performance ceiling, and the reading that the method “reaches” 0.01 is exactly the confound the protocol removes.
- 4) **A replicated null.** A variance-aware promotion criterion that gates on a confidence bound rather than the mean, designed to suppress the premature promotions that cause collapse, works no better than the plain ADR-style rule. It fails to separate on the cube ($p = 0.29$), a second object ($p = 0.88$), a second hand ($p = 0.53$), and a third hand ($p = 0.46$), and it fails even under injected estimator noise, the regime it was built for. The explanation is structural. When a demotion can undo any mistake, a

premature promotion costs a brief excursion rather than a collapse, so caution buys almost nothing and *reversibility subsumes caution*. The claim is scoped to the tested regime, where a working demotion path is always present, and it says nothing about settings where demotion is weak or absent. The null is reported with a power analysis, in the spirit of [10].

The remainder of this paper is organized as follows. Section II reviews the related work. Section III introduces the task, scheduler family, and structural account of collapse. Section IV describes the experimental protocol and statistical analysis. Section V presents the four main findings, including the dual failure mode observed across embodiments. Section VI discusses the limitations and concludes the paper.

II. RELATED WORK

Three streams of prior work provide the foundation for this study. Dexterous in-hand manipulation defines the task setting and evaluation gates examined here. Automatic curriculum learning provides the scheduler family and methodological basis. Studies of RL evaluation under small-sample conditions guide the statistical analysis.

A. Dexterous in-hand manipulation

Learning-based in-hand reorientation was established at scale by OpenAI’s Dactyl system [3] and its Rubik’s-cube successor [4], then extended to general object sets and novel shapes [5], [11], visuotactile reorientation [12], fast sim-to-real transfer [6], and demonstration-seeded auto-curricula [7]. These systems treat the curriculum as engineering plumbing, and standard benchmarks pose reorientation at loose success thresholds [13], so none report how the gate behaves at a hand’s precision limit. This work closes that gap, making the curriculum the object of study, at fixed learner and reward, at the tight end of the tolerance axis.

B. Automatic curricula

Automatic curriculum learning in RL is a mature field [1], [2], [14], spanning teacher-student algorithms [15], self-paced formulations [16], level replay [17], and unsupervised environment design [18]. A recent thread selects tasks by *learnability*, the Bernoulli success variance $p(1-p)$, maximized at $p \approx 0.5$ [19], [20]. The controller here sits at the opposite, high-reliability end of its competence axis ($\theta_{hi} = 10$), and the reason matters. At a precision frontier, pushing difficulty into the maximally-learnable band commits the schedule to rungs it cannot yet hold, which is the collapse characterized in Sec. V. Bidirectional pacing is prior art, most directly ADR [4], whose two-threshold rule the baseline reproduces on the tolerance axis. Self-paced RL [16] likewise permits regressions, and applied work still embeds success-threshold curricula as unexamined components [21], [22]. The gaps this study addresses are the collapse characterization at precision (including the fair monotone gate), the tolerance-matched protocol, the measured mechanism, and the replicated null. The scheduler is a classical control-limit policy [23], and the monotone variant deletes the corrective action.

C. Small-sample RL evaluation

Small- n deep RL comparisons are fragile, with a handful of seeds, point estimates are dominated by sampling noise and parametric test over-reject, so apparent differences frequently fail to replicate [10]. We therefore follow that program’s prescriptions, using rank-based tests, effect sizes, bootstrap intervals rather than point estimates, because our seed-level outcomes are bimodal (a run either holds the tolerance or collapses to near-zero, so a mean is uninformative across the two models). we take the collapse rate as the headline metric. To these we add the settled-tolerance column and tolerancematch grid that our specific confound demands.

III. METHODOLOGY

This section defines the task and its difficulty axis, the competence signal the scheduler reads, the seven-arm scheduler family, and a structural account of why monotone gates collapse.

A. Task and difficulty axis

The task is the stock Isaac Lab Shadow Hand reorientation environment: a 20-actuated-DoF hand holds an object (default: a 5.15 cm cube) and rotates it to a sampled goal; on success a new goal is sampled, so one 10 s episode chains many reorientations. Observations are 157-dimensional (joint state, fingertip poses, object and goal pose, previous action); actions are 60 Hz joint position targets; domain randomization perturbs friction, gains, mass, and gravity. The policy is trained with PPO [24] over 2048 parallel environments for 5000 updates (≈ 163 M frames, ≈ 36 min on one RTX 5070). Orientation error is the geodesic distance on $SO(3)$,

$$\rho(q, q_g) = 2 \arcsin(\|\text{vec}(q \otimes q_g^{-1})\|_2) \in [0, \pi], \quad (1)$$

with q, q_g unit quaternions, $\otimes$ the Hamilton product, $\text{vec}(\cdot)$ the imaginary part; the arcsin form is well conditioned near $\rho = 0$. A reorientation succeeds iff $\rho \leq \tau$. The reward is the stock shaped reward

$$r = -10 \|p - p_g\| + \frac{1}{\rho+0.1} - 2 \times 10^{-4} \|a\|^2 + 250 \cdot \mathcal{K}[\rho \leq \tau], \quad (2)$$

identical across all arms; the tolerance enters only through the sparse success indicator, so any performance difference between arms is attributable to the schedule alone. (The stock reward also carries a fall-penalty term, $f_{\text{fall}} \cdot \mathcal{K}[\|p - p_g\| \geq 0.24]$ with $f_{\text{fall}} = 0$ in our runs; a drop is instead penalized by episode termination, which forfeits the remaining success bonuses.) Difficulty is a scalar $d \in [0, 1]$ mapped to tolerance by log-interpolation,

$$\tau(d) = \tau_0 (\tau_1 / \tau_0)^d, \quad \tau_0 = 0.10, \tau_1 = 0.01, \quad (3)$$

so each step of $\delta = 0.1$ tightens the tolerance by the same factor ($\approx 20.6\%$); geometric spacing equalizes control difficulty per step, where arithmetic spacing would make the hard halving from 0.02 to 0.01 as cheap as the mild $0.10 \rightarrow 0.09$ step. Because d is continuous, a run may settle at any $\tau_{\text{final}} = \tau(d_{\text{final}})$, which we always report.

Input: mode; $\tau_0=0.10$, $\tau_1=0.01$, $\theta_{hi}=10$, $\theta_{lo}=4$, $W=5$, $m=3$, $\delta=0.1$, $K=64$, $\beta=1$

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1: every  $K$  reward calls: read  $c \leftarrow$  mean consecutive successes
2: append  $c$  to window  $H$  ( $|H| \leq W$ );  $\bar{c} \leftarrow \text{mean}(H)$ 
3:  $ready \leftarrow (|H|=W) \wedge (dwell \geq m)$ 
4:  $SE \leftarrow \text{sd}(H)/\sqrt{W}$ ;  $\ell \leftarrow \bar{c} - \beta SE$ ;  $u \leftarrow \bar{c} + \beta SE$ 
5: if  $ready \wedge \ell \geq \theta_{hi} \wedge d < 1$ :  $d \leftarrow \min(1, d + \delta)$ ;  $dwell \leftarrow 0$   $\triangleright$  promote
6: else if  $ready \wedge u < \theta_{lo} \wedge d > 0$ :  $d \leftarrow \max(0, d - \delta)$ ;  $dwell \leftarrow 0$   $\triangleright$  demote
7: apply  $\tau(d) = \tau_0(\tau_1/\tau_0)^d$  to the success test

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Algorithm 1: Reversible competence curriculum (one decision)

B. Competence signal

The environment maintains a cross-environment exponential moving average c of consecutive-success counts (smoothing factor 0.1, effective span ≈ 20 updates). The scheduler reads c every $K = 64$ reward calls and keeps a trailing window of $W = 5$ readings with mean $\bar{c}$ and standard error $SE = \hat{\sigma}/\sqrt{W}$. We gate on a success *count* rather than return because counts are comparable across tolerances (the shaping term in (2) is not) and directly measure goal-chaining; the estimator stays noisy despite two smoothing stages because successes are rare at tight tolerance, and that noise is what spuriously clears promotion thresholds.

C. The scheduler family

We keep all arms in one control-limit template so that only one structural choice, whether the gate may demote, varies. All arms share promotion threshold $\theta_{hi} = 10$, demotion threshold $\theta_{lo} = 4$ (the gap is hysteresis), minimum dwell $m = 3$, and step $\delta = 0.1$:

$$d \leftarrow \begin{cases} \min(1, d + \delta), & \ell \geq \theta_{hi} \quad (\text{promote}) \\ \max(0, d - \delta), & u < \theta_{lo} \quad (\text{demote}) \\ d, & \text{otherwise,} \end{cases} \quad (4)$$

with $\ell = \bar{c} - \beta SE$ and $u = \bar{c} + \beta SE$. The seven arms are: none ($d \equiv 1$, no curriculum); fixed ($d = \min(1, t/T)$, a feedback-free clock); adaptive (promote on a single raw reading, never demote, the de facto monotone gate); reversible (4) with $\beta = 0$, the ADR-style baseline [4]); reversible_ucb (ours, $\beta = 1$, so promotion needs the lower confidence bound to clear θ_{hi} and the bar self-tightens when readings are noisy); and two ablations, *nodenoise* (raw c , keeps demotion) and *nohyst* (de-noised, deletes demotion, the *fair monotone* gate differing from the bidirectional arms only in monotonicity). The *reversible_ucb* promotion test is a one-sided z -test rejecting “competence is only θ_{hi} ” at level $1 - \Phi(\beta) \approx 0.16$ ($\beta=0$ is the same test at level 0.5); Algorithm 1 gives the per-decision logic. The scheduler is a ~ 60 -line learner-agnostic controller writing one float (the success tolerance) into the reward with no environment rebuild; the learner is untouched and the tolerance is the sole coupling.

D. Why monotone gates collapse

The failure has a compact structural explanation. Let $c^*(d)$ be the competence the policy can sustain at difficulty d , and

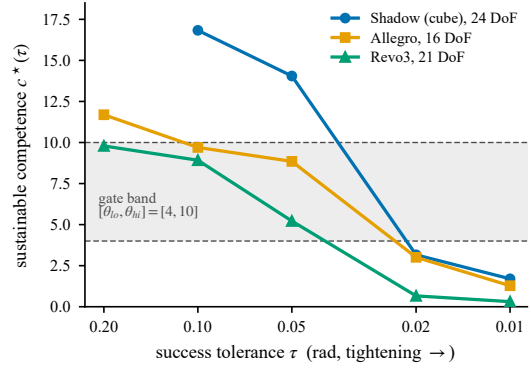


Fig. 1. A common cause, two failure modes. From-scratch sustainable competence $c^*(\tau)$ for three simulated hands, read against the fixed gate band $[\theta_{lo}, \theta_{hi}] = [4, 10]$. Each ceiling falls as τ tightens; what differs by hand is where it sits relative to the band. Shadow starts well above the band and Revo3 at its top, and both dive through the band to the cliff, so a monotone gate over-promotes onto a rung it cannot sustain (collapse); Allegro’s mid-ladder ceiling instead lingers just under θ_{hi} , so the same Shadow-calibrated gate rarely promotes (stall). A bidirectional gate instead settles where the ceiling lies inside the band. The band is Shadow-calibrated, so transplanting it unchanged is exactly what mistunes the gate on another hand.

assume (A1) c^* is non-increasing in d , (A2) when successes become rare the learning signal starves and c^* does not recover on its own, and (A3) estimator noise grows as the tolerance tightens. A2 is the load-bearing empirical assumption; the pinned no-curriculum arm, stuck over-difficulty for 85% of training (Sec. V-A), is our direct evidence for it. Under A3 a window occasionally fluctuates above θ_{hi} even when $c^*(d) < \theta_{hi}$; the gate promotes; by A1 the truth at $d + \delta$ is lower still. Model d as a Markov chain on the step grid: a monotone gate has transition support only on $\{\text{stay, up}\}$, so any state with $c^* < \theta_{lo}$ reached this way is absorbing (promotion needs $\bar{c} \geq \theta_{hi}$, which by A2 has vanishing probability, and no down-transition exists). That absorbing state is the collapse. The demotion branch adds a down-transition that fires with probability bounded away from zero exactly there, so no such state is absorbing; the chain becomes recurrent on the hysteresis band around a sustainable operating point d^* with $c^*(d^*) \in [\theta_{lo}, \theta_{hi}]$. The settled tolerance $\tau(d^*)$ is graceful degradation, not mastery of τ_1 , and the argument never references the embodiment: a hand enters only through its own c^* and noise profile (Fig. 1). This is structural motivation under stated assumptions, not a theorem.

IV. EXPERIMENTAL PROTOCOL

Within each experiment all arms share identical PPO hyperparameters, network, environment count, frame budget, and seed sets, so scheduler comparisons are causal; only the scheduler varies. The generality experiments change the object or hand and, because each embodiment was set to its own stable operating point, may use a different environment count and control rate. Therefore, reading cross-embodiment results qualitatively (which failure mode appears) rather than as a precision comparison across hands. Cube headline arms use $n = 10$ seeds because the claim is about reliability and the outcomes are bimodal (a run either masters the task or

collapses); ablation and cross-embodiment panels use $n = 3-5$ and are never pooled with them. The tolerance-matched counts are larger ($n = 14$, Table II) because that evaluation scores every available checkpoint of an arm (headline plus same-setting pilot seeds), while the robustness set is the frozen $n = 10$ block; the two are reported separately. Three metrics each close a loophole: (i) *Robustness at the hard target*: final competence, collapse rate (fraction of seeds with final $c < 3$), and the settled tolerance τ_{final} , which must be reported because a low collapse rate can be bought by backing off. (ii) *Tolerance-matched evaluation*: every run’s best-reward checkpoint (conservative, since collapsed runs keep their best pre-collapse policy) is rolled out deterministically at a common grid $\tau^* \in \{0.01, 0.02, 0.05\}$, discarding the first half of each 2500-step rollout so the competence EMA equilibrates; this removes the settled-tolerance confound. (iii) *Within-run mechanism counts*: from each run’s logs we reconstruct promotions, premature promotions ($c < \theta_{\text{lo}}$ before the next event), recoveries, and the fraction of training spent over-difficulty ($c < \theta_{\text{lo}}$ while $d > 0$), which test *why* arms differ. Outcomes are small- n and bimodal, so we use Mann–Whitney U (two-sided, tie-corrected), Cliff’s δ_{Cliff} , Holm correction, and 10k-resample bootstrap CIs [10]; the collapse rate, not the mean, is the headline. A Noether power analysis for the Mann–Whitney test puts the minimum detectable effect at $n=10$ at $\delta_{\text{Cliff}} \approx 0.72$ ($\alpha = 0.05$, power 0.8), so we write “no detectable advantage” rather than “no advantage”; the $n = 5$ cross-embodiment nulls are weaker still and replicate the direction of the cube result rather than confirm it independently.

V. EXPERIMENTS AND DISCUSSION

Two patterns recur across the results that follow: a demotion branch prevents collapse, and no scheduler masters 0.01 rad. From-scratch training at fixed tolerances confirms the difficulty cliff: final competence falls from 16.8 at $\tau=0.10$ to 3.2 at 0.02 and collapses at 0.01, the regime where a curriculum should help. The cliff is defined under the stock reward (2); a denser terminal reward could shift it, so “hard” is relative to that reward, not absolute.

A. Robustness at the hard target $\tau_1 = 0.01$

Table I is the headline experiment and Fig. 2 shows it at seed level. Reading the collapse column: no curriculum collapses 9/10; the unbuffered monotone gate 7/10; the fair de-noised monotone gate still 5/10, and at a tight settled tolerance (0.013), so smoothing alone neither prevents collapse nor is the failure a weak-baseline artifact. Every arm with a demotion branch collapses 0/10, stable across the cutoff ($c < 2/3/4$: none 8/9/9, adaptive 6/7/7, bidirectional $\leq 1/10$). Only none is measured at 0.01; the bidirectional arms avoid collapse partly by settling looser ($\tau_{\text{final}} \approx 0.020-0.027$), graceful degradation rather than mastery of 0.01 (scored fairly in Sec. V-B). The statistics support a family-level claim, not a method win: both bidirectional arms beat the fair monotone gate ($p = 0.0052/0.019$, $\delta_{\text{Cliff}} = +0.74/+0.62$) and the unbuffered baselines by more (vs.

TABLE I
ROBUSTNESS OVER $n=10$ SEEDS AT THE HARD TARGET $\tau_1=0.01$
(CUBE). COLLAPSE: FINAL $c < 3$. τ_{final} : SETTLED TOLERANCE (RAD).

scheduler	c (mean $\pm$ std)	95% CI	collapse	τ_{final}
none	1.70 ± 2.41	[0.67, 3.31]	9/10	0.010
adaptive (monotone)	3.33 ± 4.54	[0.93, 6.13]	7/10	0.014
nohyst (fair monotone)	3.58 ± 3.35	[1.69, 5.63]	5/10	0.013
nodenoise (bidir.)	8.79 ± 1.02	[8.16, 9.36]	0/10	0.022
reversible (ADR-style)	7.55 ± 2.32	[6.18, 8.89]	0/10	0.020
reversible_ucb (ours)	8.26 ± 2.19	[6.86, 9.41]	0/10	0.027

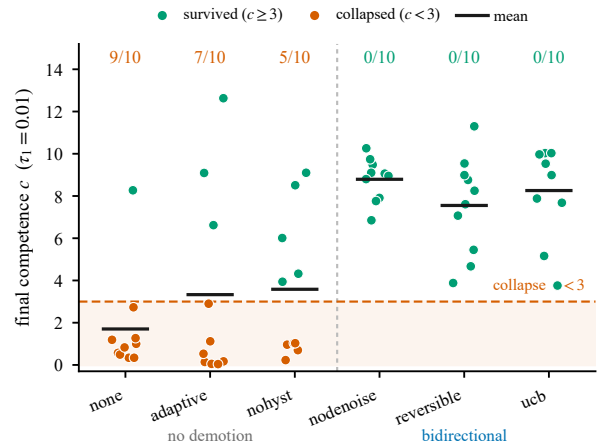


Fig. 2. The collapse drains away as reversibility is added. Per-seed final competence for all ten seeds of each arm at $\tau_1 = 0.01$; the shaded band is the collapse region ($c < 3$). Arms without a demotion branch drop 9, 7, and 5 of 10 seeds into collapse; every bidirectional arm drops none. The seed-level bimodality is why means understate the reliability gap and why the collapse rate is the fair statistic.

none: $p = 0.0005/0.0009$, $\delta_{\text{Cliff}} = +0.92/+0.88$), while reversible_ucb is indistinguishable from reversible ($p = 0.29$). The load-bearing mechanism is bidirectionality; the variance-aware refinement adds nothing detectable.

Within-run event counts turn the absorbing-state argument into statistics (mean per run, $n=10$; Fig. 3). Estimator noise is comparable across gated arms (0.22–0.28), so bidirectional arms do not win by a cleaner signal, and premature promotions occur in *every* gated arm (1.6–2.7 per run): the noisy estimate clears θ_{hi} regardless of the gate. What differs is what happens next. Monotone gates reverse none and recover none by construction, and sit over-difficulty for 23–30% of training (the pinned no-curriculum arm 85%); bidirectional arms reverse about two promotions per run and recover five to seven times, cutting over-difficulty to 1–3%.

B. Tolerance-matched evaluation

Scoring every policy at a common fixed tolerance (Table II) removes the settled-tolerance confound. At a true 0.01 the bidirectional arms lead only modestly (2.60/2.42 vs. 1.32/2.10) with overlapping spreads, and Holm-corrected tests are not significant ($p \approx 0.21$ vs. none, n.s. vs. adaptive); separation appears only at 0.02/0.05, which are not the motivating regime. The single-seed fixed clock illustrates the confound the protocol catches: it posts the table’s best score at the easy tolerance (17.49 at 0.05) precisely because a misconfigured horizon meant it never approached 0.01 (where it is worst, 0.71). Bidirectional

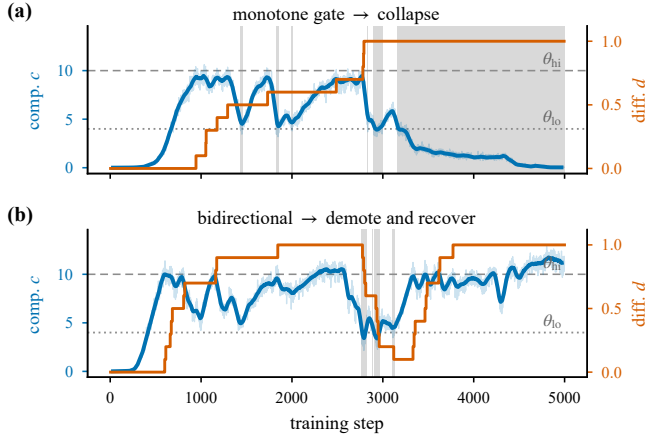


Fig. 3. The collapse mechanism in two cube runs, each the median-outcome seed of its arm (smoothed competence c in blue over the faint raw trace, left axis; difficulty d in orange, right axis; the shaded band marks over-difficulty, $c < \theta_{lo}$ while $d > 0$). (a) Monotone gate: a premature promotion ratchets difficulty to the hard target and competence starves to near zero, an absorbing state (collapse). (b) Bidirectional gate: a demotion near step 3000 lets competence recover and difficulty re-climb (graceful degradation). The gate rule is identical; only the demotion branch differs.

TABLE II

TOLERANCE-MATCHED EVALUATION (CUBE): BEST-REWARD CHECKPOINTS SCORED AT COMMON FIXED TOLERANCES τ^* (RAD); ENTRIES: c (MEAN $\pm$ STD). ABLATIONS ($n=3$) ARE A SEPARATE PANEL.

scheduler	n	$\tau^*=0.01$	$\tau^*=0.02$	$\tau^*=0.05$
none	14	1.32 ± 2.01	3.22 ± 4.14	5.76 ± 4.33
fixed	1	0.71	2.84	17.49
adaptive	14	2.10 ± 2.38	5.95 ± 4.42	9.14 ± 5.36
reversible	14	2.60 ± 1.89	7.25 ± 4.56	9.66 ± 5.60
reversible.ucb	14	2.42 ± 2.06	6.68 ± 4.71	9.59 ± 5.65
abl. nodenoise	3	4.85 ± 0.92	10.88 ± 1.39	13.44 ± 0.75
abl. nohyst	3	3.18 ± 2.60	7.63 ± 5.17	13.01 ± 1.13

gating thus buys reliable training; the ceiling at the target precision is unchanged.

C. Generality across objects

The cube result generalizes to a second geometry. On the egg ($n=10$), both bidirectional arms collapse 0/10 against 5–6/10 for every monotone arm, settling at $\bar{\tau} = 0.015$ –0.017; the collapse rate stays decisive because egg-none is bimodal (one seed solves 0.01 outright), so mean-based tests vs. none are non-significant ($p \approx 0.15$ –0.17) even though the reliability gap is stark (vs. adaptive, $p = 0.019$ –0.034). The egg tolerance-matched panel replicates the cube’s, all arms inseparable at 0.01 ($p \geq 0.096$), and the variance-aware null replicates ($p = 0.88$). The pen is different: every arm fails because the object cannot be held, so difficulty never leaves the easy end. That is a graspability barrier, not a precision one, flagged by a kill-gate that fires when an object cannot sustain competence $c \geq \theta_{lo}$ even at the loosest tolerance τ_0 , so no schedule can make progress. This is a stated exclusion criterion, not a per-object tuning choice; such objects fall outside a precision curriculum, where hold-oriented reward design [25] is the natural tool.

D. Cross-embodiment and a second failure mode

The study was repeated on an Allegro hand (16 DoF, 4 fingers) with a calibration-first protocol. From-scratch runs at fixed $\tau \in \{0.20, 0.10, 0.05, 0.02, 0.01\}$ locate the hand’s precision frontier (competence 11.69/9.70/8.85/3.01/1.28), then the arm-by-seed matrix runs there ($n = 5$). Curriculum necessity replicates with the maximum effect. No-curriculum collapses 5/5 while every gated arm collapses 0/5 (Fig. 4; $U = 0$, $p = 0.009$, $\delta_{\text{Cliff}} = +1.00$, gated means 9.2–9.7 against 0.87). The monotone-collapse mechanism went untested here, and the reason is itself the finding. Allegro’s mid-ladder ceiling (≈ 8.9 –9.7) sits just under $\theta_{hi} = 10$, so promotions were rare, two of five seeds never left the start rung, and no gated arm reached the cliff. They stalled near $\tau \approx 0.13$ –0.14 under this budget (Fig. 4), and recalibrating to $\theta_{hi} = 8$ only moved the stall one rung deeper ($\tau \approx 0.065$). A fixed scalar threshold cannot track a ceiling that falls with difficulty, because the ceiling re-crosses any recalibrated threshold further down the ladder. Monotone pacing thus has a second failure mode, under-promotion stall, dual to the over-promotion collapse on Shadow (Fig. 1). The reversible-ucb-vs.-reversible null replicated here too ($p = 0.53$).

A third embodiment (BrainCo Revo3, 21 DoF, 5 fingers; $n = 5$) closes the pattern: its easy-rung ceiling reaches the top of the band (sustained 9.8, peak 10.3 at $\tau_0=0.20$) and dives through it, so gated arms advance without stalling and reach the cliff. There, no-curriculum again collapses 5/5 (none mean 0.91) while every bidirectional arm degrades gracefully (0/5); the confidence-bound gate again only matches ADR ($p = 0.46$ on settled tolerance, $p \geq 0.92$ matched), and at a matched 0.01 rad the gated arms are indistinguishable from none (0.47–0.58). Necessity and the caution null are now three-embodiment results, where each hand settles is set by the same mechanism, its competence ceiling read against the fixed gate, rather than a shared numeric limit under these training budgets none settles at 0.01 rad.

E. Reversibility subsumes caution

The variance-aware gate followed a plausible theory. Premature promotions cause collapse, they happen when a noisy estimate clears a fixed bar, and a bar that self-tightens with the standard error should suppress them. The suppression is real, since the margin scales as βSE , but its value depends on the cost of the mistakes it prevents, and demotion makes that cost small. A false promotion costs a bounded excursion, not a run. The prediction that caution starts paying once the estimator is noisy is falsifiable, and the data falsify it. Under injected noise ($\sigma \in \{0, 3, 6\}$) the cautious gates settle looser as noise grows, reaching 0.014 (ADR), 0.022 (UCB), and 0.038 (SPRT-style) at $\sigma = 6$, because a more hesitant rule simply climbs less while demotion already caps the damage of climbing too eagerly. The five replications never separate the two gates (cube $p = 0.29$, egg $p = 0.71/0.88$, Allegro $p = 0.53$, Revo3 $p = 0.46$), and the pooled effect carries only $\approx 18\%$ power at $\delta_{\text{Cliff}} = 0.28$. What stands is a design

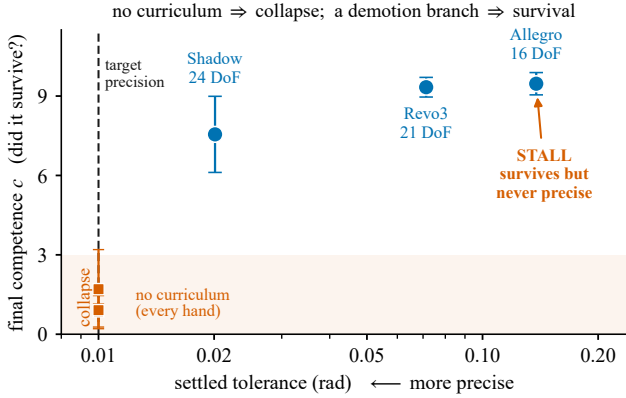


Fig. 4. Cross-embodiment outcomes in one view (real data; values computed from the canonical run tables, $n=10$ cube, $n=5$ Allegro and Revo3; whiskers are 95% CIs). The vertical axis is final competence, whether training survives, and the horizontal axis is the tolerance a run settles at, how precise it gets (log scale, tighter to the left). No-curriculum training (orange) collapses below the threshold on every hand at the pinned hard tolerance. A demotion branch (blue) lifts every hand to survival where each then settles is governed by the mechanism, the point at which its competence ceiling meets the fixed gate band. Allegro *stalls* short of the target because its mid-ladder ceiling lingers just under θ_{hi} , the under-promotion failure mode dual to over-promotion collapse (Fig. 1); Shadow and Revo3, whose ceilings dive through the band, keep tightening. The plotted settled tolerances illustrate this mechanism under each hand’s training budget, not intrinsic embodiment limits, and reading them qualitatively (stall vs. graceful reach) rather than as a precision ranking across hands.

principle, not a method. When a curriculum can undo its mistakes, caution about committing them is redundant, and effort is better spent guaranteeing a working demotion path.

VI. LIMITATIONS AND CONCLUSION

This study asked how a competence-gated curriculum behaves when pushed to a hand’s precision limit, and the answer held across every run. A monotone gate, one that can only tighten, fails there: a noisy estimate clears the promotion threshold by luck, the tolerance steps down, the success signal dies, and with no way back the run does not recover. A single demotion branch removes that failure, and training then settles reliably at the tightest tolerance the hand can hold. Gating promotion on a confidence bound rather than the mean added nothing across five replications. The mechanism is a compact, learner-agnostic controller that sets one number and leaves the learner untouched.

What the study does not show matters just as much. No scheduler tested here masters 0.01 rad. The bidirectional gate settles looser, roughly 0.020 to 0.027 rad, so it buys a reliably finished run rather than a higher precision ceiling. The tolerance a hand settles at depends on its training budget, the parallel environment count and control rate, not on the embodiment alone; we therefore treat the cross-embodiment settled tolerances as illustrations of the mechanism (which failure mode a hand exhibits) rather than as intrinsic precision limits, and do not rank hands by the tolerance they reach. The rule is ADR read on the tolerance axis, not new, and the variance-aware refinement is a confirmed negative. The evidence is two objects and three hands in one simulator with no hardware, so sim-to-real transfer is left for future work.

The pen is excluded by the kill-gate, and the absorbing-state account (Sec. III-D) is motivation, not proof.

The cross-embodiment results point to where this should go next. A fixed count threshold is mistuned the moment the hand changes, so a gate ought to read a quantity that carries across hands, a success probability held inside a target band, with a separate test for whether the object can be held at all. Whichever gate is used, report the tolerance a curriculum settled at, score every policy at the same fixed tolerance, and count the promotions and demotions the mechanism should produce.

DECLARATIONS

AI usage. Claude (Anthropic) was used to assist with manuscript editing and with writing the data-analysis and figure-generation code (Figs. 1–4) and the supplementary video. All experiments, data, numbers, and claims were produced and verified by the author(s), who take full responsibility for the paper.

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